

In this unit we do 2 very important theorems.

1. Rolle's theorem (see p. 287)
2. The Mean Value Theorem (see p. 288)

MB to read textbook!

What does it mean if $f'(c) = 0$?



Examples:

① Find c in Rolle's theorem for $f(x) = x^3 - x + 1$ on $[0, 1]$.

- ① f is continuous on $\mathbb{R} \Rightarrow f$ is continuous on $[0, 1]$
- ② f is differentiable on $\mathbb{R} \Rightarrow f$ is differentiable on $[0, 1]$
- ③ $f(0) = 1 = f(1)$

$\Rightarrow$ According to Rolle's theorem there exists a $c \in (0, 1)$, such that $f'(c) = 0$.

$$\text{Now we find } f'(x). \quad f(x) = x^3 - x + 1$$

$$\Rightarrow f'(x) = 3x^2 - 1$$

$$f'(c) = 3c^2 - 1 = 0$$

$$\Rightarrow c^2 = \frac{1}{3}$$

$$\Rightarrow \underline{c = +\frac{1}{\sqrt{3}}}, \text{ since } c \in (0, 1).$$

② Find c of the Mean Value Theorem for $f(x) = x^2 + 1$ on $[-1, 2]$.

We know ① f is continuous on $\mathbb{R}$, therefore:
 f continuous on $[-1, 2]$.

② f is differentiable on $\mathbb{R}$, i.e. f is differentiable on $(-1, 2)$.

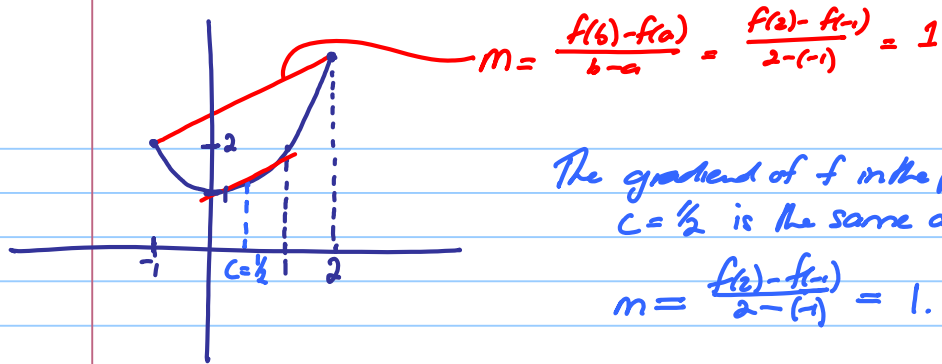
$\therefore$ According to the Mean Value Theorem, there will be a $c \in (-1, 2)$ such that

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)} = \frac{5 - 2}{3} = \frac{3}{3} = 1.$$

$$\text{Now: } f(x) = x^2 + 1 \Rightarrow f'(x) = 2x$$

$$\Rightarrow f'(c) = 2c = 1$$

$$\Rightarrow \underline{c = \frac{1}{2}}.$$



Do example 3 (p.287) and example 5 (p.290).